THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Compared with Earth, Mars has a thin atmosphere and is: 1

(a) covered by forests **(b)** very hot
(c) very cold and dry **(d)** covered by oceans

2. The region of the Earth where living organisms are found is the: 1

(a) core **(b)** atmosphere only **(c)** mantle **(d)** biosphere

3. Aeroplanes generally fly in the: 1

(a) core **(b)** thermosphere
(c) mesosphere **(d)** lower stratosphere or upper troposphere

4. Venus is much hotter than Earth because of: 1

(a) its thick carbon dioxide atmosphere **(b)** its closeness to the Moon
(c) its many oceans **(d)** its magnetic field

5. About what part of the Earth's surface is covered by water? 1

(a) about 10 per cent **(b)** about 71 per cent
(c) about 29 per cent **(d)** about 50 per cent

6. Gases that trap heat in the atmosphere are called: 1

(a) fuel gases **(b)** noble gases **(c)** greenhouse gases **(d)** inert gases

7. The Earth's rotation and tilt give us: 1

(a) day, night and seasons **(b)** volcanoes
(c) tides only **(d)** eclipses only

8. The solid rocky part of the Earth's surface is the: 1

(a) lithosphere **(b)** biosphere **(c)** atmosphere **(d)** hydrosphere

9. A greenhouse gas is: 1

(a) argon **(b)** helium **(c)** carbon dioxide **(d)** nitrogen

10. The age of the Earth is about: 1

(a) 4.5 million years **(b)** 45 thousand years
(c) 4.5 billion years **(d)** 450 years

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the Earth have day and night? 2

12. Name the layers of the atmosphere in order from the ground. 2

13. What is the greenhouse effect? 2

14. What are the effects of global warming? State two. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the greenhouse effect and name three greenhouse gases. 3

16. List three causes of global warming and three measures to control it. 3

17. Explain what the term 'sustainable use of Earth's resources' means with two examples. 3

OR

Describe three conditions that make Earth suitable for life. 3

18. Explain how the atmosphere protects us from the Sun and from meteors. 3

OR

How do the layers of the atmosphere differ in temperature trend? Give three. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the greenhouse effect, global warming and their consequences, and the measures to combat them. 5

OR

Compare the Earth's day, year and seasons with those of the Moon and Mars in terms of motions and conditions. 5

20. Describe the water on Earth: its distribution, the water cycle and the problem of fresh water. Give five points. 5

OR

Describe the conservation of the Earth's resources: water, soil, forests, air with two methods for each. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND